Statistical Analysis of Heart Failure Risk Factors

Age, Maximum Heart Rate & Cardiovascular Patterns

Author: Haneen Saeed
Institution: Multimedia University (MMU), Cyberjaya
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Dataset Overview

Dataset Title:
Heart Failure Prediction Dataset (Kaggle)

Source:
[Kaggle - Heart Failure Prediction]

Variables Selected:

Numerical Variable 1: Age
Numerical Variable 2: MaxHR (Maximum Heart Rate Achieved)
Categorical Variable: ChestPainType

Justification for Dataset Selection:

I chose this dataset because it focuses on heart disease, which is a major health issue around the world. The data includes important factors like age, maximum heart rate (MaxHR), and chest pain type, which are commonly analyzed in relation to cardiovascular health. I found this dataset interesting because it allows me to explore how these variables are connected and to see if any patterns emerge that could help explain heart disease.

Age is a well-known risk factor for many health conditions, including heart disease, while MaxHR reflects a person's cardiovascular response to physical activity and is commonly used to assess heart function. Chest pain is one of the most common symptoms in heart patients and can vary significantly among individuals.

These variables allow me to ask meaningful questions, such as how age and heart rate differ across patients and whether there's any noticeable relationship between them. I'm also curious to explore the consistency of these two numerical variables and to examine whether the age variable follows a normal distribution.

Research Questions:

1. How do Age and MaxHR vary among patients?
2. Which of the two variables (Age or MaxHR) is more consistent?
3. Does the distribution of Age follow a normal pattern?
4. Is there a relationship between Age and MaxHR?

Part A: Exploratory Descriptive Analysis

1. Summary Statistics

For Variable 1:

```
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> age <- heart_data$Age
>
> print (mean(age))
[1] 53.51089
>
> print (median(age))
[1] 54
>
> print(as.numeric(names(sort(table(heart_data$Age), decreasing = TRUE)[1])))
[1] 54
>
> print (sd(age))
[1] 9.432617
>
> print (var(age))
[1] 88.97425
>
> print (diff(range(age)))
[1] 49
>
> print (quantile(age))
  0%  25%  50%  75% 100%
  28   47   54   60   77
>
> print (IQR(age))
[1] 13
>
> cv <- (sd(age) / mean(age))*100
> print (cv)
[1] 17.62747
```

Detailed Summary:

The average age of patients in the dataset is 53.51 years, and the middle value (median) is 54 years, which are very close. This shows that the data is fairly evenly spread, without much skew. The most common age (mode) is also 54, which supports this idea.

The standard deviation is 9.43, meaning that most patients' ages are around 9 years above or below the average. The variance is 88.97, which also shows how spread out the data is.

The age range is 49 years, going from the youngest patient at 28 to the oldest at 77. The first quartile (Q1) is 47 and the third quartile (Q3) is 60, so the IQR is 13. This means that the middle 50% of ages fall between 47 and 60.

The coefficient of variation (CV) is 17.63%, which means the data isn't very spread out compared to the average. Overall, age is fairly consistent in this dataset.

For Variable 2:

```
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> HR <- heart_data$MaxHR
>
> print (mean(HR))
[1] 136.8094
>
> print (median(HR))
[1] 138
>
> print(as.numeric(names(sort(table(heart_data$MaxHR), decreasing = TRUE)[1])))
[1] 150
>
> print (sd(HR))
[1] 25.46033
>
> print (var(HR))
[1] 648.2286
>
> print (diff(range(HR)))
[1] 142
>
> print (quantile(HR))
  0%  25%  50%  75% 100%
  60  120  138  156  202
>
> print (IQR(HR))
[1] 36
>
> cv <- (sd(HR) / mean(HR))*100
> print (cv)
[1] 18.61008
```

Detailed Summary:

The average maximum heart rate (MaxHR) is 136.81 bpm, and the median is 138 bpm, which are close, suggesting the data is also fairly balanced. The most frequent MaxHR value (mode) is 150 bpm, which is higher than the mean and median. This could mean the data is slightly skewed to the left.

The standard deviation is 25.46, showing that people's MaxHR values are spread out more compared to age. The variance is 648.23, which also shows a high spread.

The MaxHR range is 142 bpm, from a low of 60 to a high of 202 bpm, showing a wide variation. The Q1 is 120 and Q3 is 156, so the IQR is 36 bpm. This means half of the patients fall in that heart rate range.

The CV is 18.61%, slightly higher than age, which tells us that MaxHR is less consistent and has more variation between individuals.

2. Comparison & Consistency

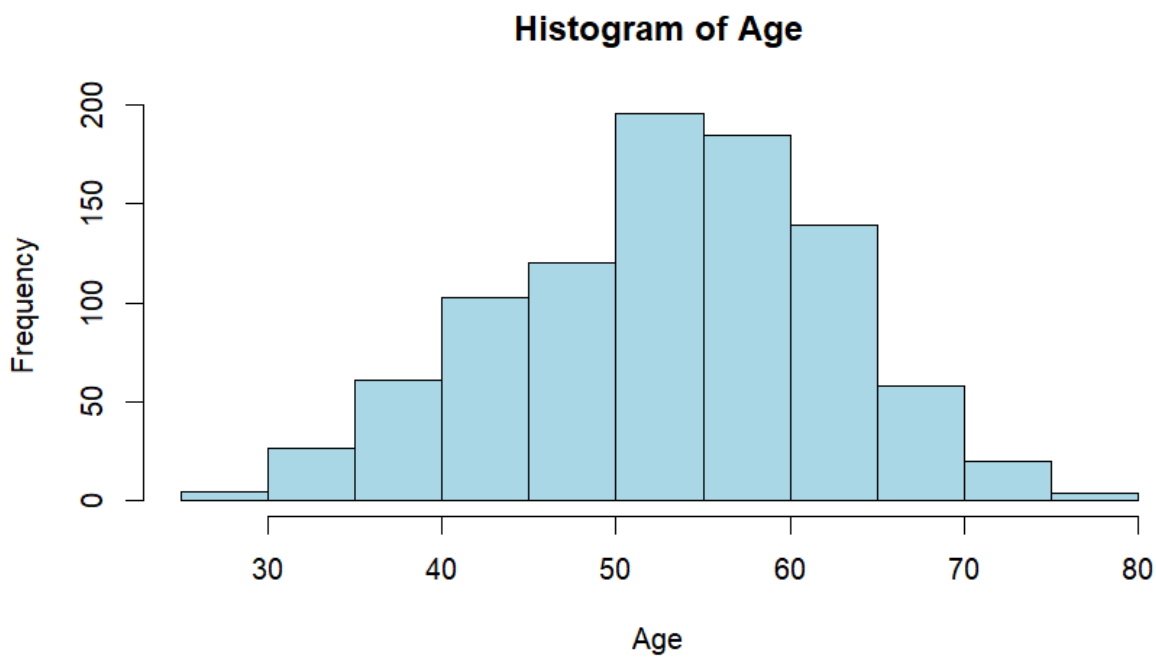
The standard deviation for Maximum Heart Rate (MaxHR) is 25.46, whereas for Age, it is 9.43. This tells us that MaxHR values in the dataset are more spread out around their mean value (136.81) compared to the spread of Age values around their mean (53.51). It clearly shows that individuals' MaxHR measurements differ more significantly from person to person than their ages do.

However, comparing standard deviations directly can be misleading when the means of the two variables are very different, which is the case here. To make a fair comparison, we use the coefficient of variation (CV). The CV for Age is approximately 17.63%, while the CV for MaxHR is 18.61%. These percentages reflect how much variability exists in each variable relative to their average values.

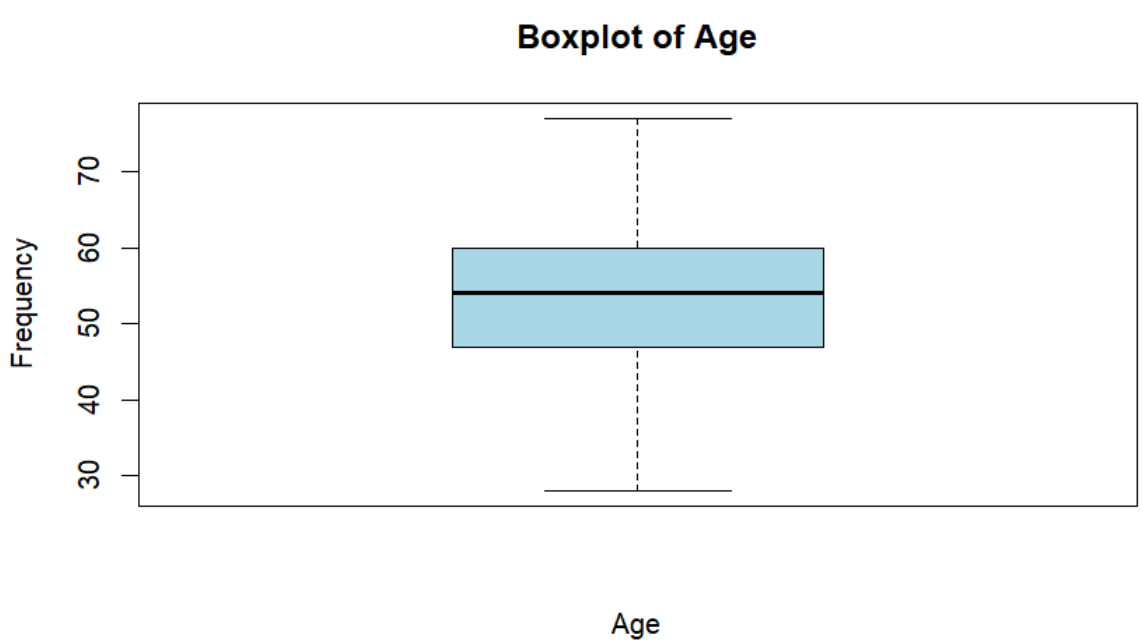
From this, we can conclude that Age is more consistent than MaxHR because it has a lower CV, meaning its values are more tightly grouped around the mean. Even though MaxHR has a larger standard deviation, its higher CV confirms that it is relatively less consistent when normalized for its scale.

3. Visualizations and Interpretation

Age :




```
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> age <- heart_data$Age
> hist(age, main = "Histogram of Age", xlab = "Age", ylab = "Frequency", col = "lightblue")
```

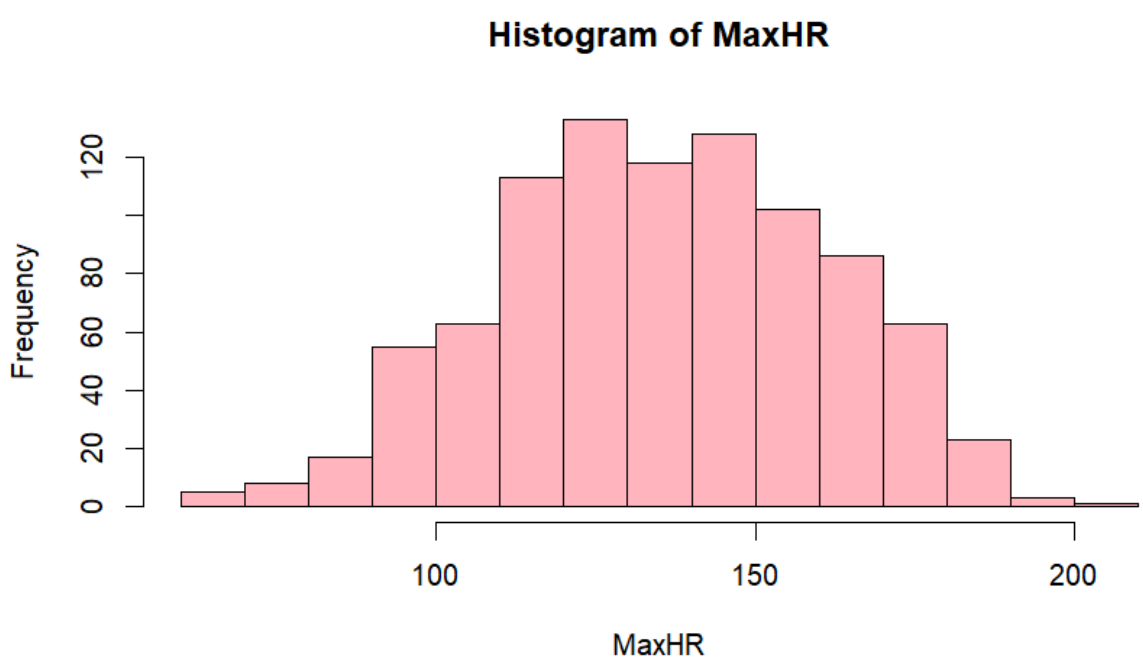


```
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> age <- heart_data$Age
> boxplot(age, main= "Boxplot of Age", xlab ="Age", ylab = "Frequency", col = "lightblue")
```

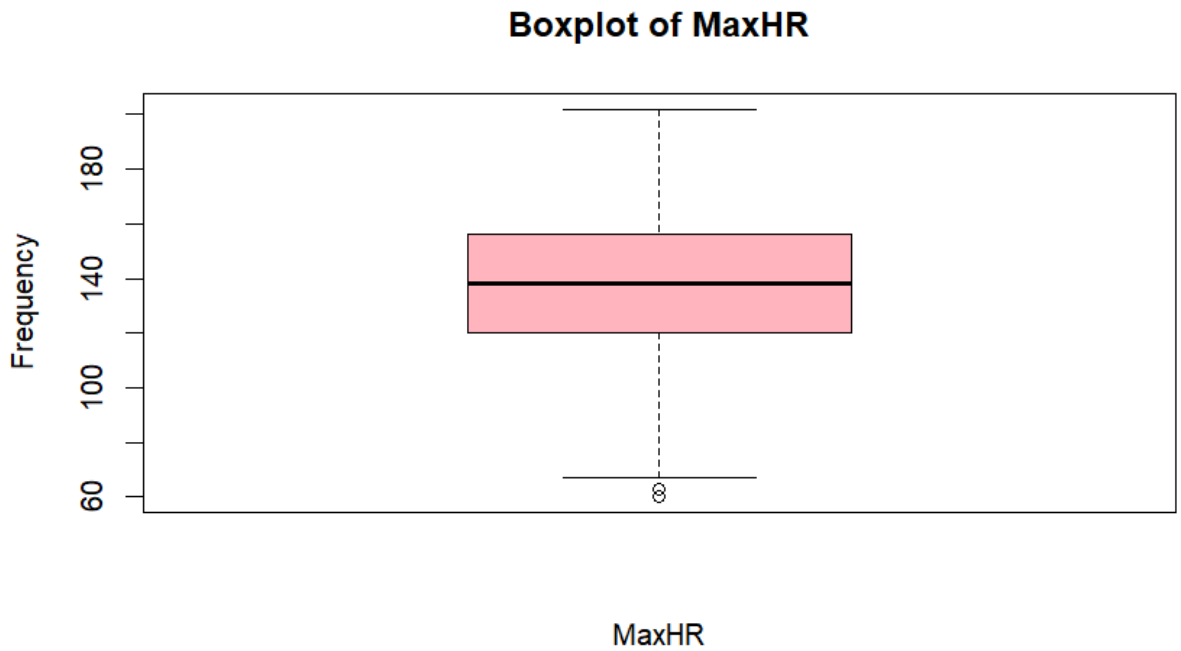
Interpretation:

Skewness:
The histogram for Age shows that the distribution is approximately symmetric, with a slight left skew. The peak is around the 50–55 age range, and the frequencies taper off gradually on both sides. While the left side has a slightly longer tail, the skewness is not very strong, so overall, the data can be considered nearly symmetric.
Outliers:
The boxplot for Age does not show any individual points outside the whiskers, which means there are no outliers in this variable. All the data values fall within a reasonable range, making the distribution quite clean and free from extreme values.
Symmetry:
Both the histogram and boxplot suggest a fairly symmetrical distribution. In the boxplot, the median line is positioned close to the center of the box, and the whiskers are of similar length. This reinforces the idea that Age is distributed evenly around the median, without any significant skew or imbalance.

MaxHR (Maximum Heart Rate)



```
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> HR <- heart_data$MaxHR
> hist(HR, main = "Histogram of MaxHR", xlab = "MaxHR", ylab = "Frequency", col = "lightpink")
```




```
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> HR <- heart_data$MaxHR
> boxplot(HR, main= "Boxplot of MaxHR", xlab ="MaxHR", ylab = "Frequency", col = "lightpink")
```

Interpretation:

Skewness:

The histogram shows a roughly symmetric distribution for MaxHR, with frequencies peaking around the middle (150-160 bpm) and tapering off equally on both sides. However, the boxplot hints at slight left skewness, as the median appears closer to the upper quartile, and the lower whisker is shorter.

Outliers:

The boxplot clearly identifies one outlier below 60 bpm, marked as an isolated point on the far left. This represents an unusually low maximum heart rate compared to the rest of the data. No outliers are present on the higher end, meaning no extreme high MaxHR values were recorded.

Symmetry:

The histogram's shape suggests near symmetry, but the boxplot's features—such as the median's slight shift toward the top half and the uneven whisker lengths—indicate minor asymmetry. The distribution is approximately normal but with a subtle leftward skew, likely due to the influence of the low outlier pulling the tail in that direction.

Part B: Distributional Assessment

Selected Variable: Age

1. Empirical Rule Check for Variable: Age

Empirical Rule:

```
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> age <- heart_data$Age
> cat ("68% range: ", round((mean(age))-(sd(age)), 2), "to", round ((mean(age))+(sd(age)),2),"\n")
68% range:  44.08 to 62.94
>
> cat ("95% range: ", round((mean(age))-2*(sd(age)), 2), "to", round ((mean(age))+2*(sd(age)),
2),"\n")
95% range:  34.65 to 72.38
>
> cat ("99.7% range: ", round((mean(age))-3*(sd(age)), 2), "to", round ((mean(age))+3*(sd(age)),
2),"\n")
99.7% range:  25.21 to 81.81
```

Ranges:

68% range: 44.08 to 62.94
95% range: 34.65 to 72.38
99.7% range: 25.21 to 81.81

The proportion of data points that fall within one, two, and three standard deviations from the mean:

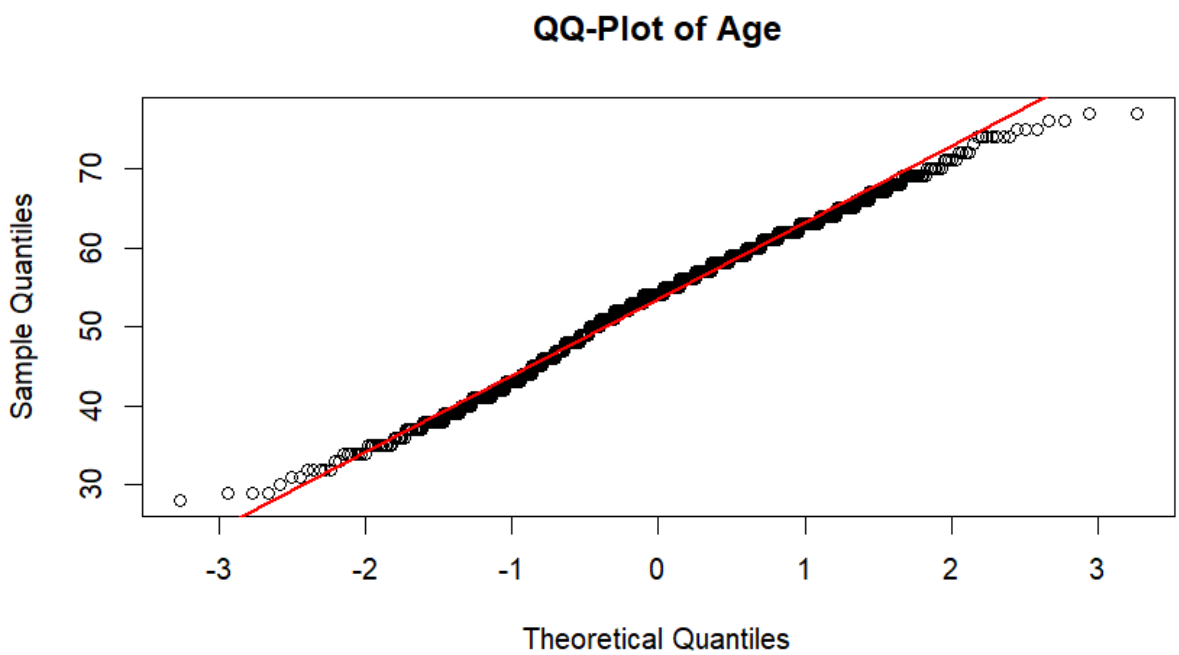
```
> mean_age <- mean(age)
> sd_age <- sd(age)
>
> within_1sd <- sum(age > (mean_age - sd_age) & age < (mean_age + sd_age)) / length(age)
> cat("Proportion within 1 SD (68%):", round(within_1sd * 100, 2), "%\n")
Proportion within 1 SD (68%): 63.73 %
>
> within_2sd <- sum(age > (mean_age - 2*sd_age) & age < (mean_age + 2*sd_age)) / length(age)
> cat("Proportion within 2 SD (95%):", round(within_2sd * 100, 2), "%\n")
Proportion within 2 SD (95%): 96.08 %
>
> within_3sd <- sum(age > (mean_age - 3*sd_age) & age < (mean_age + 3*sd_age)) / length(age)
> cat("Proportion within 3 SD (99.7%):", round(within_3sd * 100, 2), "%\n")
Proportion within 3 SD (99.7%): 100 %
```

Proportions:

Proportion within 1 SD (68%): 63.73 %
Proportion within 2 SD (95%): 96.08 %
Proportion within 3 SD (99.7%): 100 %

2. QQ-Plot and Shapiro-Wilk Test

QQ-Plot:




```
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> age <- heart_data$Age
> qqnorm(age, main = "QQ-Plot of Age", xlab = "Theoretical Quantiles", ylab = "Sample Quantiles")
> qqline(age, col = "red", lwd = 2)
```

Interpretation:

The QQ-plot for the "age" variable shows some deviation from the reference line, particularly at the tails, indicating that the data is not perfectly normally distributed. While the points near the center align relatively well with the line, the ends suggest skewness or potential outliers. This visual assessment supports the findings from the Shapiro-Wilk test, which also indicated that the "age" variable does not follow a normal distribution.

```
> shapiro_test_result <- shapiro.test(age)
> shapiro_test_result

        Shapiro-Wilk normality test

data:  age
W = 0.99101, p-value = 2.165e-05
```

Interpretation:

The test statistic was W = 0.99101, and the p-value = 2.165e-05. Since the p-value is much smaller than the significance level of 0.05, we reject the null hypothesis that the "age" variable follows a normal distribution. This suggests that the data for age is not normally distributed.This aligns with the visual assessment from the QQ-plot, where the data points showed some deviation from the expected line, indicating a departure from normality.

Normality Assessment of the "Age" Variable:

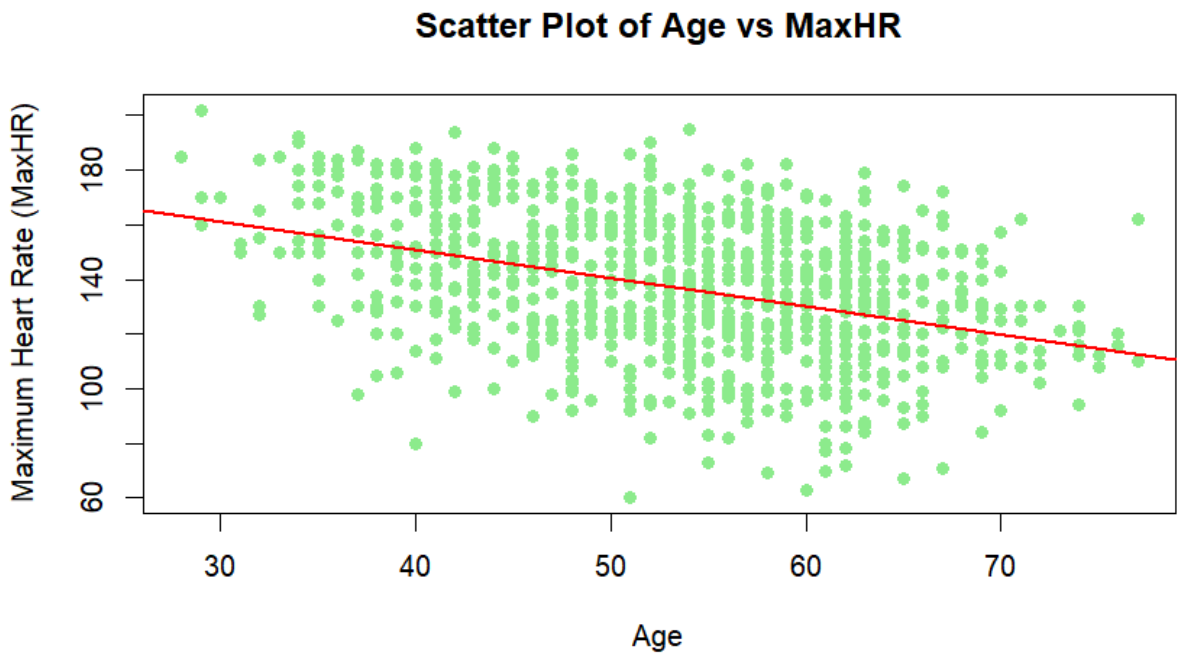
Based on the analysis, the "age" variable does not appear to follow a normal distribution. The empirical rule (68%-95%-99.7%) suggests that the data falls within expected ranges, but the proportion within the 1, 2, and 3 standard deviations from the mean was not perfectly aligned with the empirical rule.
The QQ-plot showed deviation from the reference line, especially at the tails, indicating that the distribution is not normal. Additionally, the Shapiro-Wilk test gave a significant p-value (2.165e-05), leading us to reject the null hypothesis of normality.

3. Most Reliable and Practical Method:

The Shapiro-Wilk test is the most reliable and practical method for testing normality in this case. It provides a clear statistical result and works well for small to medium-sized datasets, like this one. Although the QQ-plot offers visual insight into normality, it can be subjective, especially when deviations are not immediately obvious.
The empirical rule is a quick check but can be misleading if the data doesn't exactly follow the expected percentages. The Shapiro-Wilk test provides a definitive p-value and is a standard method for assessing normality, making it the most reliable for this analysis.

Part C: Bivariate Relationship Analysis

1. Scatter Plot with Regression Line




```
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> plot(heart_data$Age, heart_data$MaxHR,main = "Scatter Plot of Age vs MaxHR",xlab = "Age",ylab =
"Maximum Heart Rate (MaxHR)",pch = 19, col = "lightgreen")
> abline(lm(MaxHR ~ Age, data = heart_data), col = "red", lwd = 2)
```

Interpretation:

Negative Linear Relationship: The regression line slopes downward, confirming that as Age increases, MaxHR tends to decrease. This aligns with the physiological principle that maximum heart rate declines with age (e.g., the "220 - age" rough estimate).

Fit of the Regression Line: The line captures the general trend, but the moderate scatter of points around it suggests that Age alone does not fully explain MaxHR variability. Other factors (e.g., fitness, genetics, or health conditions) likely play a role.

Outliers: A few points deviate significantly, particularly one at low Age but very low MaxHR, which may indicate an unusual case (e.g., underlying heart condition or measurement error).

2. Pearson Correlation Coefficient

Step by Step:

```
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> x <- as.numeric(heart_data$Age)
>
> y <- as.numeric(heart_data$MaxHR)
>
> n <- length(x)
>
> sum_x <- sum(x)
>
> sum_y <- sum(y)
>
> sum_x2 <- sum(x^2)
>
> sum_y2 <- sum(y^2)
>
> sum_xy <- sum(x * y)
>
> Sxx <- sum_x2 - (sum_x^2 / n)
>
> Syy <- sum_y2 - (sum_y^2 / n)
>
> Sxy <- sum_xy - ((sum_x * sum_y) / n)
>
> r <- Sxy / sqrt(Sxx * Syy)
>
> r_squared <- r^2
>
> cat("Pearson Correlation Coefficient (r):", r, "\n")
Pearson Correlation Coefficient (r): -0.3820447
>
> cat("Coefficient of Determination (r²):", r_squared, "\n")
Coefficient of Determination (r²): 0.1459581
```

Pearson Correlation Coefficient (r): -0.3820447
Coefficient of Determination (r^2) : 0.1459581

Interpretation:

The calculated Pearson correlation coefficient (r = -0.38) suggests a moderate negative linear relationship between a person's age and their maximum heart rate (MaxHR). This means that, in general, as individuals grow older, their MaxHR tends to decrease. This is expected physiologically, as cardiovascular performance typically declines with age, reducing the maximum rate at which the heart can beat during physical activity.

Furthermore, the coefficient of determination ($r^2 = 0.146$) indicates that approximately 14.6% of the variation in MaxHR can be explained by age alone. While this is a significant proportion, it also highlights that other factors account for the remaining 85.4% of the variability. This includes elements like physical fitness, medication, lifestyle, and underlying health conditions, which may also impact a person's MaxHR.

In summary, while age does have a notable influence on MaxHR, it is only one part of a more complex set of factors that affect heart function. The negative correlation aligns with medical expectations and provides useful insight for understanding cardiovascular trends across age groups.

3. Causal vs Associative

The relationship between Age and MaxHR is likely associative rather than causal. The Pearson correlation indicates that as age increases, maximum heart rate tends to decrease, but this does not imply that age directly causes this change. The observed relationship could be influenced by other factors, such as a person's fitness level, health status, or lifestyle. For example, as people age, they often become less physically active, which may contribute to a lower maximum heart rate, though age itself is not necessarily the direct cause.

Part D: Application & Critical Reflection

1. Summary for a Non-Technical Audience

In this analysis, we examined the relationship between age and maximum heart rate (MaxHR) using data from a heart failure prediction dataset. Our goal was to understand how these two variables are connected. The results revealed a moderate negative relationship, which means that as a person's age increases, their maximum heart rate generally tends to decrease. This trend aligns with existing knowledge about how the cardiovascular system changes with age and how the heart's ability to respond to physical exertion declines over time.

To explore this relationship, we used statistical methods such as the Pearson correlation coefficient, which gave us a numerical value indicating the strength and direction of the relationship. We also used scatter plots with regression lines to visually represent the data, helping us see the downward trend more clearly. While age appears to be an influencing factor, other aspects like fitness level, medication, and lifestyle can also impact a person's heart rate.

Overall, this analysis helps highlight how aging may affect cardiovascular health, and shows how basic statistical tools and visualizations can provide valuable insights, even for people without a technical background.

2. Descriptive Statistics & Inferential Stats:

One limitation of descriptive statistics is that it only tells us what is happening in the sample data ,it cannot go beyond that. While it can summarize and describe the data we have, it doesn't allow us to draw conclusions about a larger population or test if our findings are meaningful. For example, even though we noticed a negative relationship between age and MaxHR, descriptive statistics alone can't confirm whether this pattern is just due to chance or if it applies more broadly.

This is where inferential statistics becomes useful. It allows us to make predictions and test hypotheses by using probability theory. By applying inferential techniques like hypothesis testing or confidence intervals, we can determine if the relationship we observed in the sample data is likely to exist in the general population as well. This adds more depth and reliability to our analysis.

3. Visualizations & Their Importance

Visualizations play a crucial role in making data more understandable, especially when dealing with large datasets or complex relationships. A graph or plot can show patterns and trends at a glance, which might be hard to spot just by looking at numbers. For instance, the scatter plot we created for age and MaxHR immediately showed a downward trend, indicating a negative relationship between the two variables.

Similarly, QQ-plots helped us assess normality more effectively than just using numerical results. Visualizations also make it easier to share findings with people who may not have a technical background, since they can convey key insights without needing to explain formulas or statistical terms. In short, they make data more accessible and meaningful.

Code in Text Form From RStudio:

Statistics For Age:

```
# Read the heart dataset into R and extract the “Age” variable
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> age <- heart_data$Age

# Calculate and print the mean of age
> print (mean(age))  # Mean of age
[1] 53.51089

# Calculate and print the median of age
> print (median(age))  # Median of age
[1] 54

# Find the most frequent age value in the dataset
> print(as.numeric(names(sort(table(heart_data$Age), decreasing = TRUE)[1])))  #
Mode of age
[1] 54

# Calculate and print the standard deviation of age
> print (sd(age))  # Standard deviation of age
[1] 9.432617

# Calculate and print the variance of age
> print (var(age))  # Variance of age
[1] 88.97425

# Calculate the range (difference between max and min) of age
> print (diff(range(age)))  # Range of age
[1] 49

# Calculate and print the quantiles of age
> print (quantile(age))  # Quantiles of age
  0%  25%  50%  75% 100%
  28   47   54   60   77

# Calculate and print the interquartile range (IQR) of age
> print (IQR(age))  # Interquartile range of age
[1] 13
```

```
# Calculate the coefficient of variation for age and print it
> cv <- (sd(age) / mean(age)) * 100  # Coefficient of variation for age
> print (cv)
[1] 17.62747
```

Statistics For MaxHR:

```
# Extract the 'MaxHR' variable from the dataset
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> HR <- heart_data$MaxHR

# Calculate and print the mean of MaxHR
> print (mean(HR))  # Mean of MaxHR
[1] 136.8094

# Calculate and print the median of MaxHR
> print (median(HR))  # Median of MaxHR
[1] 138

# Find the most frequent MaxHR value in the dataset
> print(as.numeric(names(sort(table(heart_data$MaxHR), decreasing = TRUE)[1])))  #
Mode of MaxHR
[1] 150

# Calculate and print the standard deviation of MaxHR
> print (sd(HR))  # Standard deviation of MaxHR
[1] 25.46033

# Calculate and print the variance of MaxHR
> print (var(HR))  # Variance of MaxHR
[1] 648.2286

# Calculate the range (difference between max and min) of MaxHR
> print (diff(range(HR)))  # Range of MaxHR
[1] 142

# Calculate and print the quantiles of MaxHR
> print (quantile(HR))  # Quantiles of MaxHR
  0%  25%  50%  75% 100%
  60  120  138  156  202
```

```
# Calculate and print the interquartile range (IQR) of MaxHR
> print (IQR(HR))  # Interquartile range of MaxHR
[1] 36

# Calculate the coefficient of variation for MaxHR and print it
> cv <- (sd(HR) / mean(HR)) * 100  # Coefficient of variation for MaxHR
> print (cv)
[1] 18.61008
```

Histogram for Age:

```
# Load the dataset and extract 'Age' variable again
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> age <- heart_data$Age
# Create a histogram for 'Age'
> hist(age, main = "Histogram of Age", xlab = "Age", ylab = "Frequency", col =
"lightblue")
```

Boxplot for Age:

```
# Load the dataset and extract 'Age' variable
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> age <- heart_data$Age
# Create a boxplot for 'Age'
> boxplot(age, main= "Boxplot of Age", xlab ="Age", ylab = "Frequency", col =
"lightblue")
```

Histogram for MaxHR:

```
# Load the dataset and extract 'MaxHR' variable again
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> HR <- heart_data$MaxHR
# Create a histogram for 'MaxHR'
> hist(HR, main = "Histogram of MaxHR", xlab = "MaxHR", ylab = "Frequency", col =
"lightpink")
```

Boxplot for MaxHR:

```
# Load the dataset and extract 'MaxHR' variable
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> HR <- heart_data$MaxHR
# Create a boxplot for 'MaxHR'
```

```
> boxplot(HR, main= "Boxplot of MaxHR", xlab ="MaxHR", ylab = "Frequency", col =
"lightpink")
```

Empirical Rule:

```
# Load the dataset and extract 'Age' variable
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> age <- heart_data$Age

# Calculate and display the 68% range (mean ± 1 SD)
> cat ("68% range: ", round((mean(age))-(sd(age)), 2), "to", round
((mean(age))+(sd(age)),2),"\n")

# Calculate and display the 95% range (mean ± 2 SD)
> cat ("95% range: ", round((mean(age))-2*(sd(age)), 2), "to", round
((mean(age))+2*(sd(age)),2),"\n")

# Calculate and display the 99.7% range (mean ± 3 SD)
> cat ("99.7% range: ", round((mean(age))-3*(sd(age)), 2), "to", round
((mean(age))+3*(sd(age)),2),"\n")
```

The proportion of data points that fall within one, two, and three standard deviations from the mean:

```
# Calculate mean and standard deviation for 'Age'
> mean_age <- mean(age)
> sd_age <- sd(age)

# Calculate and display the proportion within 1 SD (68%)
> within_1sd <- sum(age > (mean_age - sd_age) & age < (mean_age + sd_age)) /
length(age)
> cat("Proportion within 1 SD (68%):", round(within_1sd * 100, 2), "%\n")

# Calculate and display the proportion within 2 SD (95%)
> within_2sd <- sum(age > (mean_age - 2*sd_age) & age < (mean_age + 2*sd_age)) /
length(age)
> cat("Proportion within 2 SD (95%):", round(within_2sd * 100, 2), "%\n")

# Calculate and display the proportion within 3 SD (99.7%)
> within_3sd <- sum(age > (mean_age - 3*sd_age) & age < (mean_age + 3*sd_age)) /
length(age)
> cat("Proportion within 3 SD (99.7%):", round(within_3sd * 100, 2), "%\n")
```

QQ Plot:

```
# Load the dataset again and create a QQ-Plot for 'Age'
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> age <- heart_data$Age
> qqnorm(age, main = "QQ-Plot of Age", xlab = "Theoretical Quantiles", ylab = "Sample
Quantiles")
> qqline(age, col = "red", lwd = 2)
```

Shapiro-Wilk Test Result:

```
# Perform the Shapiro-Wilk normality test for 'Age'
> shapiro_test_result <- shapiro.test(age)
> shapiro_test_result
```

Scatter Plot Of Age vs MaxHR:

```
# Load the heart disease dataset from the specified file path and create a scatter plot
# to visualize the relationship between Age and MaxHR
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")
> plot(heart_data$Age, heart_data$MaxHR, main = "Scatter Plot of Age vs MaxHR",
    xlab = "Age", ylab = "Maximum Heart Rate (MaxHR)", pch = 19, col = "lightgreen")
# Add a regression line to the plot to show the linear relationship between Age and
MaxHR
> abline(lm(MaxHR ~ Age, data = heart_data), col = "red", lwd = 2)
```

Pearson Correlation Coefficient Step By Step:

```
# Load the dataset again and manually calculate the Pearson correlation coefficient
step-by-step
> heart_data <- read.csv ("C:/Users/Asus/Downloads/heart.csv")

> x <- as.numeric(heart_data$Age)  # Extract the Age variable
> y <- as.numeric(heart_data$MaxHR)  # Extract the MaxHR variable

# Calculate the total number of data points (n) in the dataset
> n <- length(x)

# Calculate the summation of Age (x), MaxHR (y), the squares of Age and MaxHR,
# and the product of Age and MaxHR, which are needed for Pearson correlation formula
> sum_x <- sum(x)  # Summation of all Age values
```

```
> sum_y <- sum(y)  # Summation of all MaxHR values
> sum_x2 <- sum(x^2)  # Summation of squared Age values
> sum_y2 <- sum(y^2)  # Summation of squared MaxHR values
> sum_xy <- sum(x * y)  # Summation of the product of Age and MaxHR values

# Calculate the values of Sxx, Syy, and Sxy, which are part of the Pearson correlation formula
> Sxx <- sum_x2 - (sum_x^2 / n)  # Variability of Age
> Syy <- sum_y2 - (sum_y^2 / n)  # Variability of MaxHR
> Sxy <- sum_xy - ((sum_x * sum_y) / n)  # Covariance between Age and MaxHR

# Compute the Pearson correlation coefficient (r) by dividing Sxy by the square root of Sxx and Syy
> r <- Sxy / sqrt(Sxx * Syy)

# Calculate the coefficient of determination (r²), which indicates the proportion of the variance in MaxHR that can be explained by Age
> r_squared <- r^2

# Print the calculated Pearson correlation coefficient (r) to the console
> cat("Pearson Correlation Coefficient (r):", r, "\n")

# Print the coefficient of determination (r²) to the console
> cat("Coefficient of Determination (r²):", r_squared, "\n")
```